

WALMART SALES PERFORMANCE DASHBOARD

Using Microsoft Power BI – Project Report

Submitted By: Amirthavarshini
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Tool Used: Microsoft Power BI Desktop

INTRODUCTION:

For this project, I developed a Sales Performance Dashboard using Microsoft Power BI and Walmart's weekly sales dataset. I selected this dataset because it contains different types of information such as store-wise sales, dates, fuel prices, CPI, unemployment rates, and holiday details. This provided an opportunity to study sales performance using different types of Power BI visualizations.

The main aim of creating this dashboard was to convert the raw and unorganized sales data into a clear and interactive report. Instead of viewing a large amount of information directly from the dataset, the dashboard makes it easier for a user or manager to understand important sales information quickly.

The dashboard allows users to apply filters based on factors such as store, date, and holiday status. When a filter is selected, the related charts and figures change automatically according to the selected information.

Visualizations are useful for sales analysis because they make large amounts of data easier to understand. They help identify patterns, compare different stores, observe changes over time, and understand relationships between different variables.

In this project, the visuals are connected with each other. Therefore, selecting a particular store, date, or holiday status automatically updates the related charts. This makes the dashboard interactive and more useful than a collection of separate static charts. The first visual provides an overall understanding of the sales data and acts as a starting point for analysing the performance in greater detail.

OBJECTIVE:

The main objectives that I planned to achieve through this Power BI dashboard were:

1. To analyse the weekly sales performance of Walmart across all 45 stores and understand the overall sales pattern.
2. To identify important Key Performance Indicators (KPIs), such as total sales, average weekly sales, and transaction count.
3. To develop an interactive dashboard where users can select required variables and apply different filters.
4. To compare holiday and non-holiday weeks and understand whether sales behaviour changes during these periods.
5. To study the movement of sales over different periods of time.
6. To examine whether fuel prices have any relationship with weekly sales.
7. To display the information through connected visualizations so that important trends and comparisons can be understood easily.

DATASET DESCRIPTION:

For this project, I worked with Walmart's weekly sales dataset. The dataset contains sales information collected from different Walmart stores.

The major fields used in the analysis were:

- Store: Represents the store identification number, ranging from 1 to 45.
- Date: Shows the week on which the sales were recorded.
- Weekly Sales: Represents the total sales generated by a particular store during a week.
- Fuel Price: Shows the average fuel price for the corresponding week.
- CPI: Represents the Consumer Price Index.
- Unemployment: Indicates the unemployment rate.
- Holiday Flag: Shows whether the particular week contained a holiday.

These fields provided the information required to study store performance, time-based trends, holiday patterns, and the relationship between fuel prices and sales.

DATA PREPARATION:

Before creating the dashboard visuals, I first prepared and cleaned the dataset so that Power BI could process the information correctly.

One of the main issues I noticed during the initial import was related to the Date column. It was not being recognized correctly as a date field. Therefore, I corrected the date format using Power Query.

The main data preparation steps were:

- Imported the Excel dataset into Power BI Desktop.
- Corrected the Date column and converted it into the appropriate date format.
- Checked the data types of Weekly Sales, Fuel Price, CPI, and Unemployment.
- Made sure that the numerical fields were recognized correctly.
- Created the required DAX measures for the KPI cards.

Proper data preparation was important because incorrect data types could affect calculations and visualizations in the dashboard.

DAX MEASURE:

To display important information in the KPI section, I created four main measures in Power BI:

- Total Sales
- Average Sales
- Store Count
- Maximum Sales

These measures were created using basic DAX functions such as SUM, AVERAGE, DISTINCTCOUNT, and MAX. The measures helped convert the raw Weekly Sales information into meaningful summary figures. They also made the dashboard easier to read because the most important values could be displayed at the top of the report.

DATA COMPONENTS AND VISUALS USED:

After preparing the Walmart dataset, I created different visual components in Power BI to analyse the sales performance.

Each visual was selected according to the type of information that needed to be presented. I also created measures for the required calculations and connected the visuals so that they respond when filters are applied. The major visual components used in the dashboard are explained below.

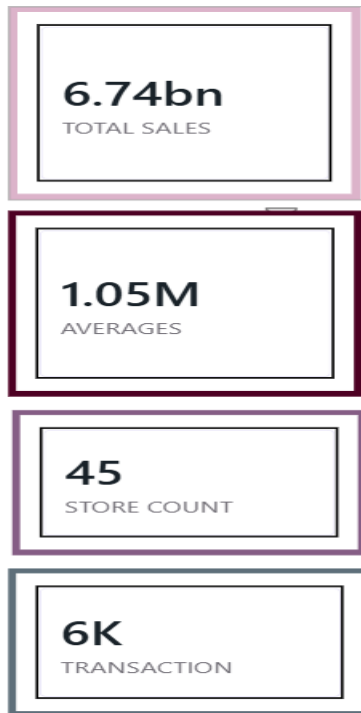
1. KPI CARDS:

The first component I created was the KPI card section. These cards provide a quick overview of the most important figures in the dataset.

The following measures were used:

- Total Sales: Shows the overall weekly sales value.
- Average Sales: Displays the average sales generated during a week.
- Transaction Count: Shows the total number of transactions.
- Store Count: Represents the number of stores included in the analysis.

I placed each measure in a separate card. This allows the user to understand the major performance indicators immediately without having to examine every individual chart. KPI cards are useful because they provide a quick summary before moving into detailed analysis.



2. SUMMARY TABLE:

I also created a summary table to examine the performance of individual stores. In this table, I used Store as the main field and included:

- Total Sales
- Average Sales
- Transaction Count

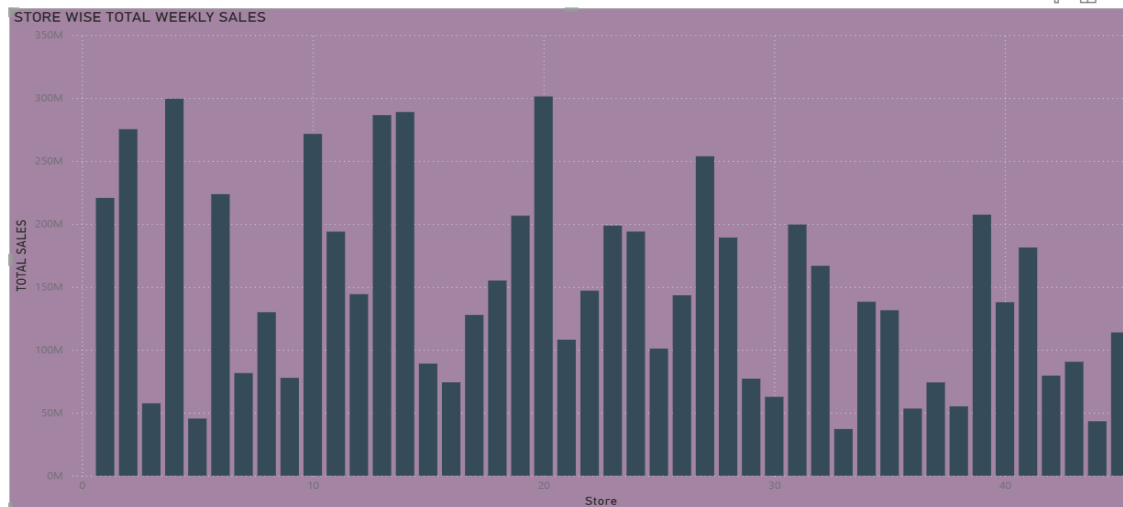
The purpose of this table was to provide more detailed store-level information. While the KPI cards provide an overall summary, the table makes it possible to compare the performance of individual stores. This helps identify stores that perform relatively better or lower in terms of sales.

Store	TOTAL SALES	AVERAGES	STORE COUNT
1	220,753,420.41	1,543,730.21	1
2	275,382,440.98	1,925,751.34	1
3	57,586,735.07	402,704.44	1
4	299,543,953.38	2,094,712.96	1
5	45,475,688.90	318,011.81	1
6	223,756,130.64	1,564,728.19	1
7	81,598,275.14	570,617.31	1
8	129,951,181.13	908,749.52	1
9	77,789,218.99	543,980.55	1
10	271,617,713.89	1,899,424.57	1
11	193,962,786.80	1,356,383.12	1
12	144,287,230.15	1,009,001.61	1
13	286,517,703.80	2,003,620.31	1
14	288,999,911.34	2,020,978.40	1
15	89,133,683.92	623,312.47	1
16	74,252,425.40	519,247.73	1
17	127,782,138.83	893,581.39	1
18	155,114,734.21	1,084,718.42	1
19	206,634,862.10	1,444,999.04	1
20	301,397,792.46	2,107,676.87	1
21	108,117,878.92	756,069.08	1
22	147,075,648.57	1,028,501.04	1
23	198,750,617.85	1,389,864.46	1
24	194,016,021.28	1,356,755.39	1
25	101,061,179.17	706,721.53	1
26	143,416,393.79	1,002,911.84	1
27	253,855,916.88	1,775,216.20	1
28	189,263,680.58	1,323,522.24	1
29	77,141,554.31	539,451.43	1
30	62,716,885.12	438,579.62	1
Total	6,737,048,005.63	1,046,613.02	45

3. TOTAL WEEKLY SALES BY STORE:

To compare the total sales generated by different Walmart stores, I created a Clustered Column Chart. I placed Store on the axis and used Total Weekly Sales as the value.

This chart provides a visual comparison between the stores. By looking at the height of each column, it becomes easier to identify stores with comparatively higher and lower total sales. This type of chart is suitable because it makes differences between individual categories easy to observe.

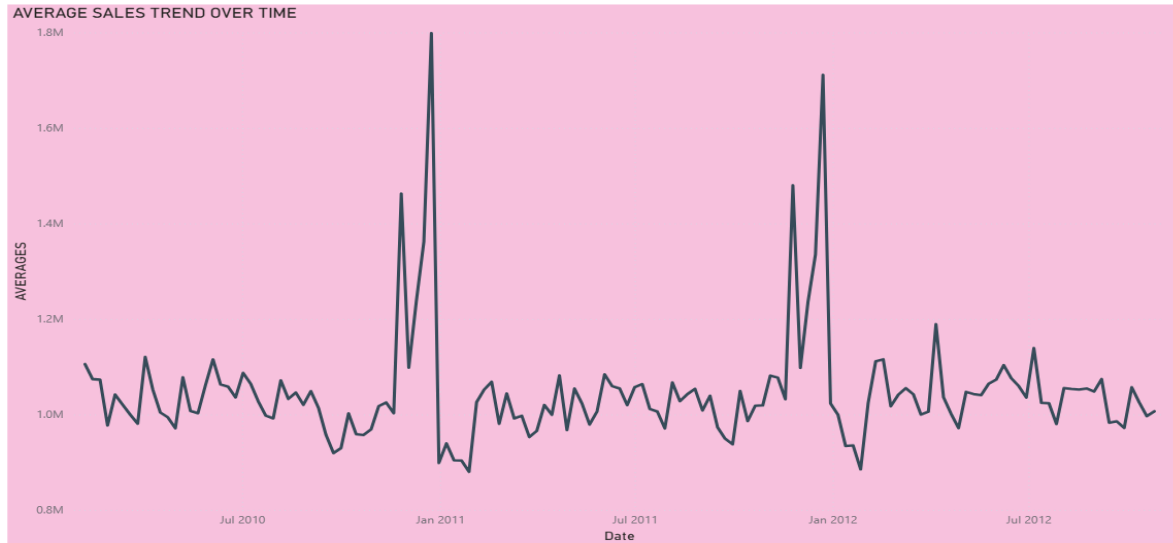


4. AVERAGE SALES TREND OVER TIME:

To study how sales changed over time, I created a Line Chart. The Date hierarchy was used on the time axis, while Average Sales was used as the value. The Date hierarchy allows the information to be viewed at different levels, including:

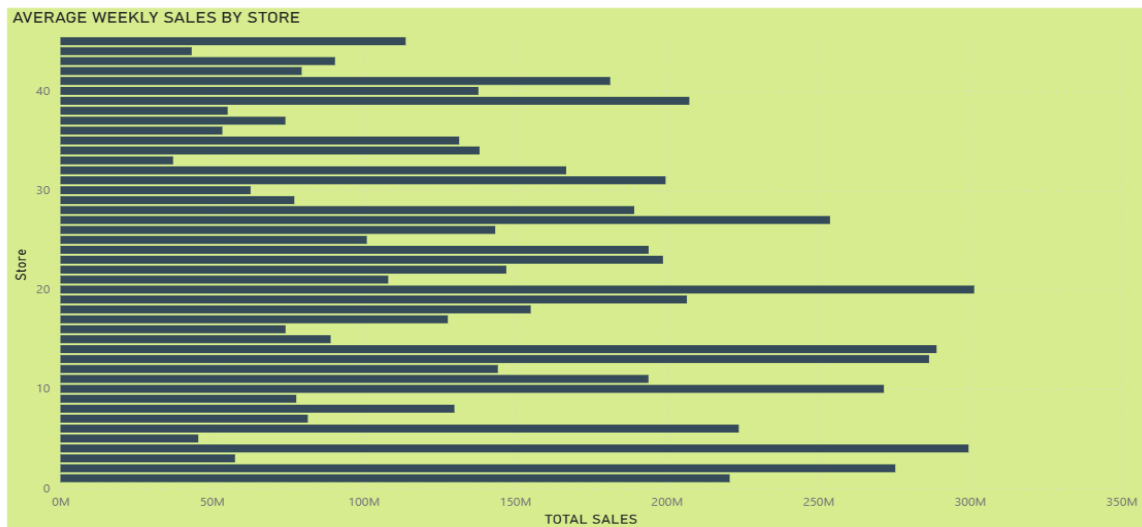
- Year
- Quarter
- Month
- Day

This visualization helps identify changes and movements in average sales across the selected period. A line chart was appropriate for this purpose because it clearly shows the direction and movement of values over time.



5. AVERAGE SALES BY STORE:

Another visualization I created was a Clustered Bar Chart to compare the average sales of different stores. I used Store as the category and Average Sales as the value. This visual provides a different perspective from total sales. Instead of only looking at the overall sales generated by each store, it allows me to understand the average sales performance of the stores. It therefore helps identify stores that have comparatively high or low average sales.



6. CUMULATIVE WEEKLY TREND:

To display the movement of sales over time, I created an Area Chart titled “CUMULATIVE WEEKLY TREND.”

For this chart, I used the Date hierarchy along with Total Weekly Sales.

I selected an area chart because it provides a clear representation of how the sales values change across the selected time period. The filled area makes it easier to observe the overall movement and trend in the sales data. This visual helps provide a broader understanding of sales performance throughout the period.

7. FUEL PRICE VS WEEKLY SALES:

I created a Scatter Chart titled “FUEL PRICE VS WEEKLY SALES” to examine whether there was any relationship between fuel prices and Walmart’s weekly sales.

For this visualization:

- Fuel Price was used as one variable.
- Total Weekly Sales was used as the other variable.

A scatter chart was selected because it is useful for comparing two numerical variables and observing whether any noticeable relationship exists between them. This visual helped me understand whether changes in fuel prices appeared to have an impact on weekly sales.

8. HOLIDAYS VS NON-HOLIDAY:

I created a Scatter Chart titled “FUEL PRICE VS WEEKLY SALES” to examine whether there was any relationship between fuel prices and Walmart’s weekly sales.

For this visualization:

- Fuel Price was used as one variable.
- Total Weekly Sales was used as the other variable.

A scatter chart was selected because it is useful for comparing two numerical variables and observing whether any noticeable relationship exists between them. This visual helped me understand whether changes in fuel prices appeared to have an impact on weekly sales.

9. STORE RANKING OVER TIME:

I created a Scatter Chart titled “FUEL PRICE VS WEEKLY SALES” to examine whether there was any relationship between fuel prices and Walmart’s weekly sales.

For this visualization:

- Fuel Price was used as one variable.
- Total Weekly Sales was used as the other variable.

A scatter chart was selected because it is useful for comparing two numerical variables and observing whether any noticeable relationship exists between them. This visual helped me understand whether changes in fuel prices appeared to have an impact on weekly sales.

10. SLICERS AND INTERACTIVITY:

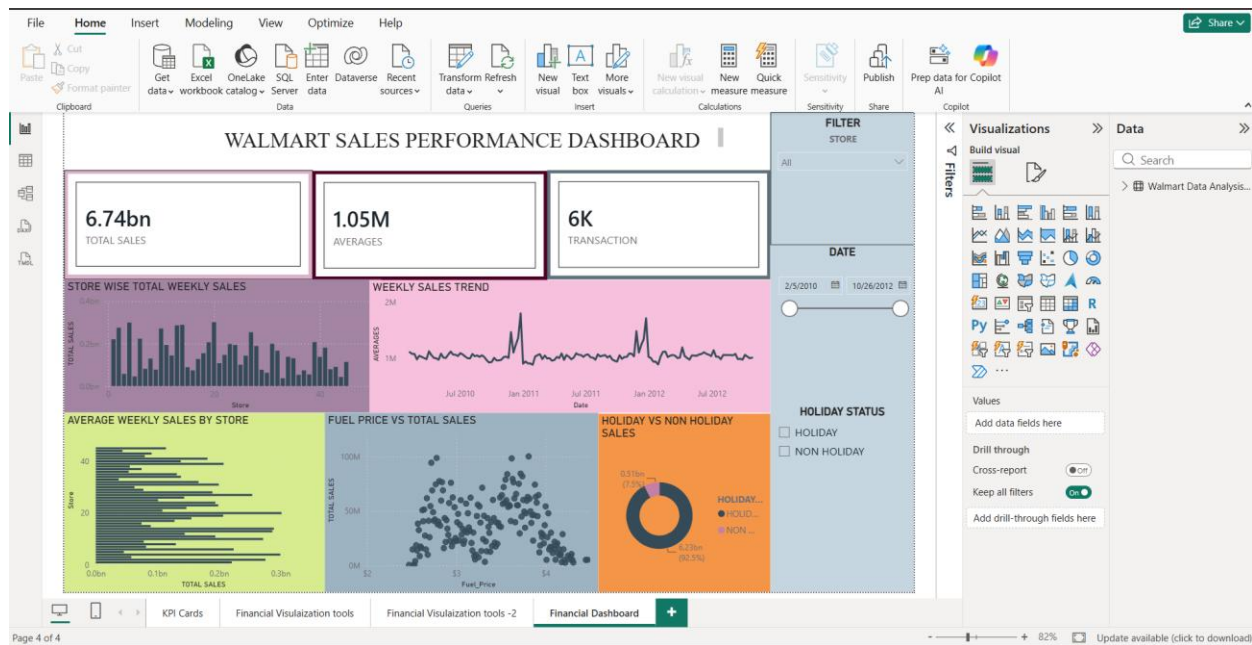
To make the dashboard more interactive, I added three slicers:

- Store: Allows the user to select a particular store.
- Date: Allows the user to choose a specific date or time period.
- Holiday Flag: Allows the user to filter the information according to holiday status.

The main benefit of using these slicers is that the visuals and KPI cards are connected.

For example, if I select one particular store, the KPI cards and charts automatically change to display information related to that store. Similarly, selecting a specific date or holiday status updates the related visuals.

This makes the dashboard interactive and allows the same dataset to be analysed from different perspectives. Overall, the visual components helped me move from a basic summary of the dataset to a more detailed analysis involving stores, sales trends, fuel prices, holidays, and store rankings.



CONCLUSION:

Through this project, I gained practical knowledge of how Microsoft Power BI can be used to transform raw sales data into an interactive and understandable dashboard.

I created KPI cards, tables, charts, and slicers to examine Walmart's weekly sales from different viewpoints. The dashboard allowed me to study store-wise performance, average sales, sales trends over time, holiday and non-holiday patterns, fuel price relationships, and changes in store rankings.

Working with visualizations made it easier to identify patterns and compare large amounts of information than simply viewing the raw dataset.

Overall, this project helped me improve my practical understanding of Power BI, particularly in selecting suitable visualizations, creating DAX measures, applying filters, and connecting different dashboard components.

It also showed me how an interactive dashboard can support data analysis by presenting complex information in a simple and understandable form. This type of dashboard can help users and managers quickly interpret important sales information and make better decisions based on the available data.